

Cardiopulmonary & Cerebral Resuscitation MCQ – Chapter 41

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 9: Cardiorespiratory Care (Chapter 41)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Cardiac arrest in adults is usually of which origin?

- A. Cardiac
- B. Metabolic
- C. Neurological
- D. Respiratory

Ans: A

Q2. Unwitnessed cardiac arrest in children should be considered due to which rhythm until proved otherwise?

- A. Pulseless electrical activity
- B. Asystole
- C. Ventricular tachycardia
- D. Ventricular fibrillation

Ans: B

Q3. Every minute delay in initiating CPR decreases prognosis by what percentage?

- A. 7–10%
- B. 5–7%
- C. 1–3%
- D. 15–20%

Ans: A

Q4. The standard CPR sequence was changed from A-B-C to C-A-B in which year's guidelines?

- A. 2015
- B. 2020
- C. 2005
- D. 2010

Ans: D

Q5. According to newer studies, what is the more important cause of airway obstruction in unconscious patients?

- A. Epiglottitis
- B. Laryngospasm
- C. Foreign body
- D. Tongue fall

Ans: A

Q6. In a patient with suspected cervical spine fracture, what is the preferred method of airway management?

- A. Jaw thrust
- B. Neck extension
- C. Head tilt and chin lift
- D. Cricothyroidotomy

Ans: A

Q7. Which device, being usable by lay rescuers, is part of Basic Life Support? A. Transcutaneous pacemaker B. Synchronized cardioverter C. Manual defibrillator D. Automatic external defibrillator (AED)	Ans: D
Q8. Mouth to mouth rescue breathing is recommended for whom? A. Only medical doctors B. Only paramedics C. Only trained rescuers D. All lay rescuers	Ans: C
Q9. What is the most definitive method of airway management during CPR? A. Laryngeal mask airway B. Guedel airway C. Combitube D. Endotracheal tube	Ans: D
Q10. What is the Heimlich maneuver used for? A. Cervical spine stabilization B. Foreign body airway obstruction in adults and older children C. Defibrillation technique D. Cardiac massage in infants	Ans: B
Q11. For cervical spine fracture patients, which type of neck stabilization is preferred during intubation? A. Manual stabilization B. Halo vest C. Hard cervical collar D. Soft cervical collar	Ans: A
Q12. During cardiac massage, the sternum should be depressed by how much? A. 2 inches (5 cm) but not more than 2.4 inches (6 cm) B. 1 inch (2.5 cm) C. 3 inches (7.5 cm) D. 1.5 inches (4 cm)	Ans: A
Q13. What is the recommended rate of chest compressions during CPR? A. 120–140/minute B. 80–100/minute C. 100–120/minute D. 60–80/minute	Ans: C

Q14. Effective cardiac output generated by successful cardiac massage is what percentage of normal? A. 30% B. 50% C. 10% D. 20%	Ans: A
Q15. What is the compression to ventilation ratio for adults WITHOUT advanced airway, irrespective of the number of rescuers? A. 15:2 B. 15:1 C. 30:2 D. 5:1	Ans: C
Q16. With advanced airway in place, what is the ventilation rate during CPR? A. 8 breaths/minute B. 10 breaths/minute C. 6 breaths/minute D. 12 breaths/minute	Ans: B
Q17. What is the minimum ETCO₂ that should be produced by successful cardiac massage? A. 20 mm Hg B. 15 mm Hg C. 10 mm Hg D. 30 mm Hg	Ans: A
Q18. ETCO₂ becoming normal (40 mm Hg) during CPR is the earliest indicator of what? A. Adequate ventilation B. Myocardial recovery C. Return of normal circulation D. Brain death	Ans: C
Q19. Which of the following is NOT a shockable rhythm? A. Asystole B. Ventricular fibrillation C. Polymorphic ventricular tachycardia (Torsade de Pointes) D. Pulseless ventricular tachycardia	Ans: A
Q20. What is the recommended energy for monophasic defibrillation? A. 120 Joules B. 200 Joules C. 300 Joules D. 360 Joules	Ans: D

Q21. What is the recommended defibrillation paddle size for adults? A. 4.5 cm B. 13 cm C. 15 cm D. 8 cm	Ans: B
Q22. After delivering a defibrillation shock, what should be done immediately? A. Check the rhythm immediately B. Give 5 cycles of 30:2 or 2 minutes of CPR before checking rhythm C. Administer adrenaline immediately D. Wait for 5 minutes and reassess	Ans: B
Q23. In monitored in-hospital patients, defibrillation should ideally be done within how many minutes? A. 3 minutes B. 5 minutes C. 1 minute D. 10 minutes	Ans: A
Q24. What is the most common cause of pulseless electrical activity (PEA)? A. Tension pneumothorax B. Hypovolemia C. Hypoxia D. Hypothermia	Ans: B
Q25. Which of the following is NOT one of the 5H causes of PEA? A. Hydrogen ion (acidosis) B. Hypovolemia C. Hypothermia D. Hypernatremia	Ans: D
Q26. Which of the following is one of the 5T causes of PEA? A. Tracheomalacia B. Thyrotoxicosis C. Tamponade (cardiac) D. Tachycardia	Ans: C
Q27. Asystole is described as: A. The terminal event of all arrhythmias with very poor prognosis B. The most common initial rhythm in adults C. The rhythm most responsive to adrenaline D. A shockable rhythm	Ans: A

Q28. Cough CPR is applicable in which situation? A. Pediatric patient with PEA B. Pregnant patient with cardiac arrest C. Conscious patient having ventricular fibrillation D. Unconscious patient with asystole	Ans: C
Q29. What is the target core body temperature range in targeted temperature management (TTM) after cardiac arrest? A. 36–38°C for 48 hours B. 32–34°C for 12 hours C. 32–36°C for 24 hours D. 28–30°C for 12 hours	Ans: C
Q30. What is the target SpO2 range in post-cardiac arrest care? A. 88–92% B. 90–94% C. 98–100% D. 94–98%	Ans: D
Q31. What blood sugar range (mg/dL) should be maintained in post-resuscitation care? A. 180–220 mg/dL B. 144–180 mg/dL C. 80–110 mg/dL D. 110–140 mg/dL	Ans: B
Q32. Emergency PCI for STEMI patients after cardiac arrest should preferably be performed within what time? A. 30 minutes B. 90 minutes C. 60 minutes D. 120 minutes	Ans: B
Q33. When should neurologic prognostication be performed after ROSC in patients NOT given TTM? A. 24 hours B. 48 hours C. 96 hours D. 72 hours	Ans: D
Q34. From a CPR point of view, what age group defines an 'infant'? A. 1 year to 5 years B. 1 year till puberty C. 4 weeks to 1 year D. Birth to 4 weeks	Ans: C

Q35. What is the compression to ventilation ratio for a CHILD with TWO rescuers and NO advanced airway? A. 3:1 B. 5:1 C. 30:2 D. 15:2	Ans: D
Q36. What is the pulse check site for infants during CPR? A. Brachial/Femoral B. Radial C. Dorsalis pedis D. Carotid	Ans: A
Q37. What heart rate in a pediatric patient with signs of poor perfusion is an indication to start chest compressions? A. <80/min B. <40/min C. <100/min D. <60/min	Ans: D
Q38. What is the first shock energy for pediatric manual defibrillation? A. 1 J/kg B. 10 J/kg C. 4 J/kg D. 2 J/kg	Ans: D
Q39. What is the compression to ventilation ratio for NEWBORN resuscitation? A. 15:2 B. 30:2 C. 3:1 D. 5:1	Ans: C
Q40. What is the indication for starting chest compressions in a newborn? A. HR <60/min despite adequate ventilation with 100% O2 for 30 seconds B. HR <100/min despite adequate ventilation C. HR <80/min despite adequate ventilation D. HR <40/min despite adequate ventilation	Ans: A
Q41. During CPR in pregnancy, how should aortocaval compression be relieved? A. Trendelenburg position B. Left lateral tilt C. Right lateral tilt D. Manual uterine displacement	Ans: D

Q42. Emergency cesarean section during maternal cardiac arrest should be considered if fetus is >25 weeks and can be performed within how many minutes? A. 10 minutes B. 2 minutes C. 4 minutes D. 6 minutes	Ans: C
Q43. What is the fluid of choice during CPR? A. Dextrose-saline B. 5% dextrose C. Normal saline D. Ringer lactate	Ans: C
Q44. What is the dose of adrenaline by endotracheal route compared to intravenous dose? A. 1.5 times IV dose B. 5 times IV dose C. Same as IV dose D. 2.5 times IV dose	Ans: D
Q45. Which of the following drugs can NOT be administered via endotracheal route? A. Lignocaine B. Adrenaline C. Amiodarone D. Naloxone	Ans: C
Q46. Routine use of sodium bicarbonate without proven acidosis during CPR is harmful because of all EXCEPT: A. Is hypertonic and increases plasma osmolarity B. Causes direct myocardial depression C. Produces CO₂ which can worsen neuronal injury D. Can cause paradoxical intracellular acidosis	Ans: B
Q47. The only recommended modality for global cerebral ischemia during CPR is: A. Barbiturates (Thiopentone) B. Targeted temperature management C. Calcium channel blockers D. Propofol infusion	Ans: B
Q48. In the criteria for brain death, no spontaneous respiration should occur even if pCO₂ increases to: A. 50 mm Hg B. 60 mm Hg C. 70 mm Hg D. 40 mm Hg	Ans: B

Q49. What is the IV dose of adrenaline during CPR and how frequently is it repeated?

- A. 5 mg once only
- B. 1 mg every 3–5 minutes
- C. 0.5 mg every 1–2 minutes
- D. 2 mg every 5–10 minutes

Ans: B

Q50. What is the first dose of amiodarone during CPR for ventricular tachycardia?

- A. 150 mg IV
- B. 300 mg IV
- C. 450 mg IV
- D. 100 mg IV

Ans: B

Answer Key & Explanations

Q1. Answer: A — Cardiac

As per the textbook, cardiac arrest in adults is usually of cardiac origin while in children it is usually of respiratory origin.

Topic: CPR Basics | Ref: Ch 41, Section: CPR Basics, Page 297

Q2. Answer: B — Asystole

Unwitnessed cardiac arrest in children should be considered due to asystole until proved otherwise, while in adults it should be considered due to ventricular fibrillation.

Topic: CPR Basics | Ref: Ch 41, Section: CPR Basics, Page 297

Q3. Answer: A — 7–10%

As per AHA chain of survival, every minute delay in CPR decreases prognosis by 7–10%.

Topic: Chain of Survival | Ref: Ch 41, Section: Management of CPR, Page 297

Q4. Answer: D — 2010

The standard sequence A (Airway) → B (Breathing) → C (Circulation) was changed to C → A → B in the 2010 CPR guidelines.

Topic: CPR Sequence | Ref: Ch 41, Section: Management of CPR, Page 297

Q5. Answer: A — Epiglottis

As per newer studies, epiglottis rather than tongue fall has been found to be the more important cause of airway obstruction in unconscious patients.

Topic: Airway Management | Ref: Ch 41, Section: Airway Management, Page 298

Q6. Answer: A — Jaw thrust

In patients with suspected cervical spine fracture, head tilt and chin lift should be avoided and airway should be managed only by jaw thrust. However, if airway is not manageable by jaw thrust alone, head tilt and chin lift can be given as life takes priority.

Topic: Airway Management | Ref: Ch 41, Section: Airway Management, Page 298

Q7. Answer: D — Automatic external defibrillator (AED)

AED can be used by lay rescuers and therefore is part of basic life support. AEDs automatically detect rhythm and deliver shock if rhythm is shockable.

Topic: Basic Life Support | Ref: Ch 41, Section: Basic Life Support, Page 298

Q8. Answer: C — Only trained rescuers

Rescue breath by mouth to mouth is reserved only for trained rescuers (medical as well as non-medical). For untrained lay rescuers, CPR means 'hands only' i.e. only cardiac massage.

Topic: Breathing | Ref: Ch 41, Section: Breathing (Ventilation), Page 300

Q9. Answer: D — Endotracheal tube

Endotracheal intubation is the most definitive and best method for ventilation during CPR.

Topic: Airway Management | Ref: Ch 41, Section: Breathing (Ventilation), Page 300

Q10. Answer: B — Foreign body airway obstruction in adults and older children

Heimlich maneuver involves abdominal compression 6–10 times with the patient standing and rescuer behind. It is used for adults and older children with foreign body airway obstruction.

Topic: Airway Management | Ref: Ch 41, Section: Management of Airway Obstruction due to Foreign Body, Page 299

Q11. Answer: A — Manual stabilization

As intubation often requires removal of cervical collar, manual stabilization is always preferred over collar stabilization.

Topic: Cervical Spine Management | Ref: Ch 41, Section: Airway and Breathing for Patients with Cervical Spine, Page 300

Q12. Answer: A — 2 inches (5 cm) but not more than 2.4 inches (6 cm)

The force generated during massage should be able to depress sternum by 2 inches (5 cm) but not more than 2.4 inches (6 cm) to avoid complications due to excessive pressure.

Topic: Circulation | Ref: Ch 41, Section: Circulation, Page 300

Q13. Answer: C — 100–120/minute

The rate of compression should be 100–120 compressions/minute but not more than 120, because at rates >120/min, compression depth decreases significantly.

Topic: Circulation | Ref: Ch 41, Section: Circulation, Page 301

Q14. Answer: A — 30%

Effective cardiac output generated by successful massage is only 30% of normal, therefore all efforts should be directed to establish spontaneous circulation.

Topic: Circulation | Ref: Ch 41, Section: Physiological Considerations of Cardiac Massage, Page 301

Q15. Answer: C — 30:2

Without advanced airway, the ratio for adults should be 30:2 (30 compressions followed by 2 breaths) irrespective of number of resuscitators.

Topic: Compression to Ventilation Ratio | Ref: Ch 41, Section: Compression to Ventilation Ratio, Page 301

Q16. Answer: B — 10 breaths/minute

With advanced airway, breathing is at a rate of 10 breaths/minute (1 breath every 6 seconds) without any synchronization with compressions.

Topic: Compression to Ventilation Ratio | Ref: Ch 41, Section: Compression to Ventilation Ratio, Page 301

Q17. Answer: A — 20 mm Hg

Successful cardiac massage should be able to produce ETCO₂ of at least 20 mm Hg. If ETCO₂ is less than 10 mm Hg, cardiac massage should be considered grossly abnormal.

Topic: Monitoring of CPR | Ref: Ch 41, Section: Monitoring of CPR, Page 301

Q18. Answer: C — Return of normal circulation

ETCO₂ becoming normal (40 mm Hg) is the earliest indicator of return of normal circulation during CPR.

Topic: Monitoring of CPR | Ref: Ch 41, Section: Monitoring of CPR, Page 301

Q19. Answer: A — Asystole

Shockable rhythms include VF, pulseless VT and polymorphic VT (Torsade de Pointes). Non-shockable rhythms include asystole and pulseless electrical activity (PEA).

Topic: Arrhythmia Management | Ref: Ch 41, Section: Management of Arrhythmias Seen During Cardiac Arrest, Page 301

Q20. Answer: D — 360 Joules

With monophasic defibrillators, all shocks should be of 360 Joules. With biphasic defibrillators, 150–200 J (exponential waveform) or 120 J (rectilinear waveform).

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

Q21. Answer: B — 13 cm

Paddle sizes: Adults – 13 cm, Children – 8 cm, Infants – 4.5 cm.

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

Q22. Answer: B — Give 5 cycles of 30:2 or 2 minutes of CPR before checking rhythm

Immediately after giving shock, give 5 cycles of 30:2 (without advanced airway) or 2 minutes of CPR (with advanced airway) before checking rhythm, as it takes ~2 minutes for heart to generate cardiac output even if rhythm becomes sinus.

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

Q23. Answer: A — 3 minutes

Ideally, in monitored patients, defibrillation should be done within 3 minutes.

Topic: Defibrillation | Ref: Ch 41, Section: Defibrillation, Page 302

Q24. Answer: B — Hypovolemia

Hypovolemia (shock) is the most common cause of PEA as stated in the textbook.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Pulseless Electrical Activity (PEA), Page 303

Q25. Answer: D — Hypernatremia

The 5H causes of PEA are: Hypovolemia, Hypokalemia/Hyperkalemia, Hypothermia, Hypoxia, and Hydrogen ion (acidosis). Hypernatremia is not included.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Causes of PEA, Page 304

Q26. Answer: C — Tamponade (cardiac)

The 5T causes of PEA are: Tension Pneumothorax, Tamponade (cardiac), Thrombosis (coronary/pulmonary), Tablets (drug overdoses)/Toxins, and Trauma.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Causes of PEA, Page 304

Q27. Answer: A — The terminal event of all arrhythmias with very poor prognosis

Asystole is the terminal event of all arrhythmias. The prognosis of asystole is very poor as compared to PEA and ventricular fibrillation.

Topic: Nonshockable Rhythms | Ref: Ch 41, Section: Asystole, Page 304

Q28. Answer: C — Conscious patient having ventricular fibrillation

Cough CPR is applicable to a conscious patient having VF. Coughing during VF increases intrathoracic pressure and can maintain cerebral perfusion (consciousness) for 90 seconds.

Topic: Newer Techniques in CPR | Ref: Ch 41, Section: Newer Techniques/Variations in CPR, Page 305

Q29. Answer: C — 32–36°C for 24 hours

TTM means maintaining core body temperature between 32–36°C for 24 hours. This was changed from the 2010 guideline of 32–34°C for 12 hours.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

Q30. Answer: D — 94–98%

Hyperoxia can produce complications including ARDS, therefore use minimum FiO₂ which maintains target SpO₂ between 94 and 98%.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

Q31. Answer: B — 144–180 mg/dL

Moderate blood sugar of 144–180 mg/dL (8–10 mmol) should be maintained in the post-resuscitation period, as both hypoglycemia and hyperglycemia are unacceptable.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

Q32. Answer: B — 90 minutes

Emergency (preferably within 90 minutes) coronary angiography/angioplasty is recommended for all patients with STEMI.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

Q33. Answer: D — 72 hours

Prognostication is recommended 72 hours after return of spontaneous circulation for those who have not been given TTM.

Topic: Post Cardiac Arrest Care | Ref: Ch 41, Section: Postcardiac Arrest Care, Page 305

Q34. Answer: C — 4 weeks to 1 year

From CPR point of view: Neonate = first 4 weeks, Infant = 4 weeks to 1 year, Child = 1 year till puberty, Adult = beyond puberty.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 305

Q35. Answer: D — 15:2

For children without advanced airway, the ratio is 30:2 for one rescuer and 15:2 for two rescuers (while for adults it is 30:2 irrespective of rescuers).

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

Q36. Answer: A — Brachial/Femoral

Pulse check sites: Infants – Brachial/Femoral, Children – Carotid, Adults – Carotid/Femoral.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

Q37. Answer: D — <60/min

Heart rate <60/min with signs of poor perfusion (pallor/cyanosis) is the indication to start chest compressions in the pediatric age group.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

Q38. Answer: D — 2 J/kg

Energy selection for shock (manual) in pediatric patients is 2 J/kg for first shock and 4 J/kg subsequently.

Topic: Pediatric CPR | Ref: Ch 41, Section: Pediatric CPR, Page 306

Q39. Answer: C — 3:1

Compression ventilation ratio for newborns is 3:1 (90 compressions with 30 breaths) and is synchronized.

Topic: Newborn CPR | Ref: Ch 41, Section: Newborn CPR, Page 306

Q40. Answer: A — HR <60/min despite adequate ventilation with 100% O2 for 30 seconds

In newborns, indication for chest compression is HR <60/min in spite of adequate ventilation with 100% oxygen for 30 seconds.

Topic: Newborn CPR | Ref: Ch 41, Section: Newborn CPR, Page 306

Q41. Answer: D — Manual uterine displacement

Lateral tilt has been eliminated from new CPR guidelines as it compromises CPR quality. Aortocaval compression should be achieved by uterine displacement.

Topic: CPR in Pregnancy | Ref: Ch 41, Section: CPR in Pregnancy, Page 307

Q42. Answer: C — 4 minutes

Emergency LSCS should be considered if fetus >25 weeks and LSCS can be performed within 4 minutes of maternal arrest.

Topic: CPR in Pregnancy | Ref: Ch 41, Section: CPR in Pregnancy, Page 307

Q43. Answer: C — Normal saline

Normal saline is the fluid of choice during CPR. Glucose-containing solutions increase cerebral edema, and Ringer lactate should be avoided as calcium present can cause neuronal injury.

Topic: Important Considerations in CPR | Ref: Ch 41, Section: Fluid During CPR, Page 307

Q44. Answer: D — 2.5 times IV dose

The dose by endotracheal route is 2.5 times of intravenous dose and the drug should be diluted with 10 mL of saline in adults and 5 mL in children.

Topic: Important Considerations in CPR | Ref: Ch 41, Section: Endotracheal (ET), Page 307

Q45. Answer: C — Amiodarone

Drugs that can be delivered by ET route are: Adrenaline, Atropine, Lignocaine (Xylocard), and Naloxone. Amiodarone is not in this list.

Topic: Important Considerations in CPR | Ref: Ch 41, Section: Endotracheal (ET), Page 307

Q46. Answer: B — Causes direct myocardial depression

The textbook lists harms of sodium bicarbonate as: CO₂ production worsening neuronal injury, increased intracranial tension, paradoxical intracellular acidosis, and hypertonic effect increasing plasma osmolarity. Direct myocardial depression is not mentioned.

Topic: Sodium Bicarbonate in CPR | Ref: Ch 41, Section: Use of Sodium Bicarbonate, Page 308

Q47. Answer: B — Targeted temperature management

The only recommended modality for global cerebral ischemia seen in CPR is targeted temperature management. Barbiturates, Propofol, and calcium channel blockers are useful for focal ischemia but their role in global ischemia is not proven.

Topic: Cerebral Protection | Ref: Ch 41, Section: Cerebral Protection, Page 308

Q48. Answer: B — 60 mm Hg

One of the clinical tests for brain death is no spontaneous respiration even if pCO₂ increases to 60 mm Hg.

Topic: Brain Death | Ref: Ch 41, Section: Brain Death, Page 308

Q49. Answer: B — 1 mg every 3–5 minutes

The CPR dose of adrenaline is 1 mg IV to be repeated every 3–5 minutes in concentration of 1:10,000.

Topic: Drugs for CPR | Ref: Ch 41, Section: Adrenaline (Epinephrine), Page 308

Q50. Answer: B — 300 mg IV

Amiodarone is the drug of choice for ventricular tachycardia. The first dose is 300 mg IV and subsequent doses are 150 mg.

Topic: Drugs for CPR | Ref: Ch 41, Section: Amiodarone, Page 308